

PDAC GWAS QC Decision Tree

Raw Genotypes (1461 samples, 430000 variants)

- > Sample Call Rate (`--mind 0.02`): remove samples with >2% missing
- > Sex Check (chrX F-stat): remove sex-discordant samples
- > Heterozygosity (F $\pm$ 3 SD): remove contamination/outlier samples
- > Variant Call Rate (`--geno 0.05`): remove variants with >5% missing
- > Hardy-Weinberg ($p > 1e-6$ in controls): remove error-prone variants
- > Relatedness (KING > 0.1875 + PI_HAT report): phenotype-aware pruning
- > MAF Filter (`--maf 0.01`): retain rare variants (PDAC strategy)

Clean Dataset Ready for Association Testing (1239 samples, 401909 variants)